

COMPILER DESIGN ASSIGNMENT 2

PES2UG23CS311

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LEXER.L

```
%option noyywrap
%option yylineno

%{
#define YYSTYPE char*

#include <stdio.h>
#include <string.h>
#include "parser.tab.h"

extern void yyerror(const char *);
%}

digit      [0-9]
letter     [a-zA-Z]
id         {letter}({letter}|{digit})*
digits     {digit}+
number     {digits}

%%

"if"       { return IF; }
"else"     { return ELSE; }
"do"       { return DO; }
"while"    { return WHILE; }

"=="       { return EQ; }
"!="       { return NE; }
">="       { return GE; }
"<="       { return LE; }
">"        { return '>'; }
```

```

"<"      { return '<'; }

"("      { return '('; }
")"      { return ')'; }
"{"      { return '{'; }
"}"      { return '}'; }
";"      { return ';'; }
"="      { return '='; }

"+"      { return '+'; }
"-"      { return '-'; }
"*"      { return '*'; }
"/"      { return '/'; }

{number} {
    yylval = strdup(yytext);
    return T_NUM;
}

{id} {
    yylval = strdup(yytext);
    return T_ID;
}

[\\t\\n ]+ ;

. ;

%%

```

PARSER.Y

```

%{
#include "quad_generation.h"
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

```

```

#define YYSTYPE char*

void yyerror(char* s);
int yylex();
extern int yylineno;

FILE* icg_quad_file;
int temp_no = 1;
int label_no = 1;

char* new_label() {
    char* label = malloc(10);
    sprintf(label, "L%d", label_no++);
    return label;
}
%}

%token T_ID T_NUM
%token IF ELSE DO WHILE
%token EQ NE GE LE

%start S

%%

S : STMT S
  | STMT
  ;

STMT
  : ASSGN ';'
  | IF '(' C ')' '{' S '}' ELSE '{' S '}'
  {
    char* L1 = new_label();
    char* L2 = new_label();

    quad_code_gen("ifFalse", $3, "goto", L1);
    quad_code_gen("", "", "label", L2);
  }
  | DO '{' S '}' WHILE '(' C ')' ';'
  {

```

```

        char* L1 = new_label();
        quad_code_gen("", "", "label", L1);
        quad_code_gen("if", $6, "goto", L1);
    }
;

ASSGN : T_ID '=' E
    {
        quad_code_gen(strdup($1), $3, "=", "");
    }
;

C : T_ID rel T_ID
    {
        $$ = new_temp();
        quad_code_gen($$, $1, $2, $3);
    }
;

rel : '<'    { $$ = "<"; }
    | '>'    { $$ = ">"; }
    | LE     { $$ = "<="; }
    | GE     { $$ = ">="; }
    | EQ     { $$ = "=="; }
    | NE     { $$ = "!="; }
;

E : E '+' T { $$ = new_temp(); quad_code_gen($$, $1, "+", $3); }
  | E '-' T { $$ = new_temp(); quad_code_gen($$, $1, "-", $3); }
  | T       { $$ = $1; }
;

T : T '*' F { $$ = new_temp(); quad_code_gen($$, $1, "*", $3); }
  | T '/' F { $$ = new_temp(); quad_code_gen($$, $1, "/", $3); }
  | F       { $$ = $1; }
;

F : '(' E ')' { $$ = $2; }
  | T_ID      { $$ = strdup($1); }
  | T_NUM     { $$ = strdup($1); }
;

```

```

%%

void yyerror(char* s)
{
    printf("Error: %s at line %d\n", s, yylineno);
}

int main()
{
    icg_quad_file = fopen("icg_quad.txt","w");
    yyparse();
    fclose(icg_quad_file);
    return 0;
}

```

QUAD_GENERATION.H

```

#ifndef QUAD_GEN_H
#define QUAD_GEN_H

#include <stdio.h>

extern FILE* icg_quad_file;
extern int temp_no;

char* new_temp();
void quad_code_gen(char* result, char* arg1, char* op, char* arg2);

#endif

```

QUAD_GENERATION.C

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include "quad_generation.h"

extern FILE* icg_quad_file;

```

```
extern int temp_no;

char* new_temp()
{
    char* temp = malloc(10);
    sprintf(temp, "t%d", temp_no++);
    return temp;
}

void quad_code_gen(char* result, char* arg1, char* op, char* arg2)
{
    printf("( %s, %s, %s, %s )\n", op, arg1, arg2, result);
    fprintf(icg_quad_file, "( %s, %s, %s, %s )\n", op, arg1, arg2, result);
}
```

TEST_INPUT_1.C

```
x = 9/2 + a - b;
```

TEST_INPUT_2.C

```
if (a < b) {
    x = a + b;
} else {
    x = a - b;
}
```

TEST_INPUT_3.C

```
do {
```

```
x = x + 1;
} while (a < b);
```

OUTPUT:

```
[maanyabhardwaj@Maanyabhardwaj ~ % cd ~/Downloads/A2
[maanyabhardwaj@Maanyabhardwaj A2 % /opt/homebrew/opt/flex/bin/flex lexer.l
maanyabhardwaj@Maanyabhardwaj A2 % /opt/homebrew/opt/bison/bin/bison -d parser.y
maanyabhardwaj@Maanyabhardwaj A2 % gcc parser.tab.c lex.yy.c quad_generation.c -o A2
[maanyabhardwaj@Maanyabhardwaj A2 % ./A2 < test_input_1.c
( /, 9, 2, t1 )
[( +, t1, a, t2 )
( -, t2, b, t3 )
( =, t3, , x )
maanyabhardwaj@Maanyabhardwaj A2 % ./A2 < test_input_2.c
( <, a, b, t1 )
[( +, a, b, t2 )
( =, t2, , x )
( -, a, b, t3 )
( =, t3, , x )
( goto, t1, L1, ifFalse )
( label, , L2, )
maanyabhardwaj@Maanyabhardwaj A2 % ./A2 < test_input_3.c
( +, x, 1, t1 )
[( =, t1, , x )
( <, a, b, t2 )
( label, , L1, )
( goto, 1, L1, if )
maanyabhardwaj@Maanyabhardwaj A2 %
```